AI and Blockchain Governance :

Enhancing Fairness in Token Voting Systems

BGIN Block #11 | Washington D.C, USA

Joseph Beverley (SingularityNET); Jessica Mila Schutzman (Advisor)

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Background

Joseph Beverley
Strategic Initiatives
SingularityNET

'Avid BGINer'

Jessica Mila Schutzman Esq., CTCE, CIPP/E
Independent Advisor
Ex-Kraken | Ex-Electric Coin Company

'BGIN OG'

What we are doing → why you should care

Minimal scholarly work has addressed the potential **impact of artificial intelligence in blockchain governance structures.**

This paper avoids attempting to evenly discuss the nuanced complexities and multifaceted nature of blockchain governance and decentralised governance as a whole.

However, by initiating our research we have in parallel identified limited consensus concerning decision making events within blockchain lack of taxonomy and agreed that if thoroughly researched and readily available would provide this paper and future papers focusing on AI and decentralised governance significant enhancements.

Blockchain

Blockchain

Super Cool

Artificial Intelligence

Artificial Intelligence

Even Cooler

Governance

Governance

Not so cool

Governance

Not so cool :(

Blockchain

Distributed Ledger

Interoperability

Proof of Stake

Decentralized Finance

Mining

Tokenization

Decentralized Autonomous Organizations

Cryptography

Transparency

Security

Smart Contracts

Decentralized Networks

Consensus Mechanisms

Immutability

Artificial Intelligence

AI-driven Innovation

AI Utility

Reinforcement Learning

Automation

Autonomous Systems

Machine Learning

Neural Networks

Predictive Analytics

Natural Language Processing

Computer Vision

AI Governance

AI Ethics

Deep Learning

Artificial General Intelligence

Data Processing

Governance

Autonomy

Incentives

Governance Models

Decision-making

Fairness

Stakeholders

Voting

Plutocracy

Consensus

Representation

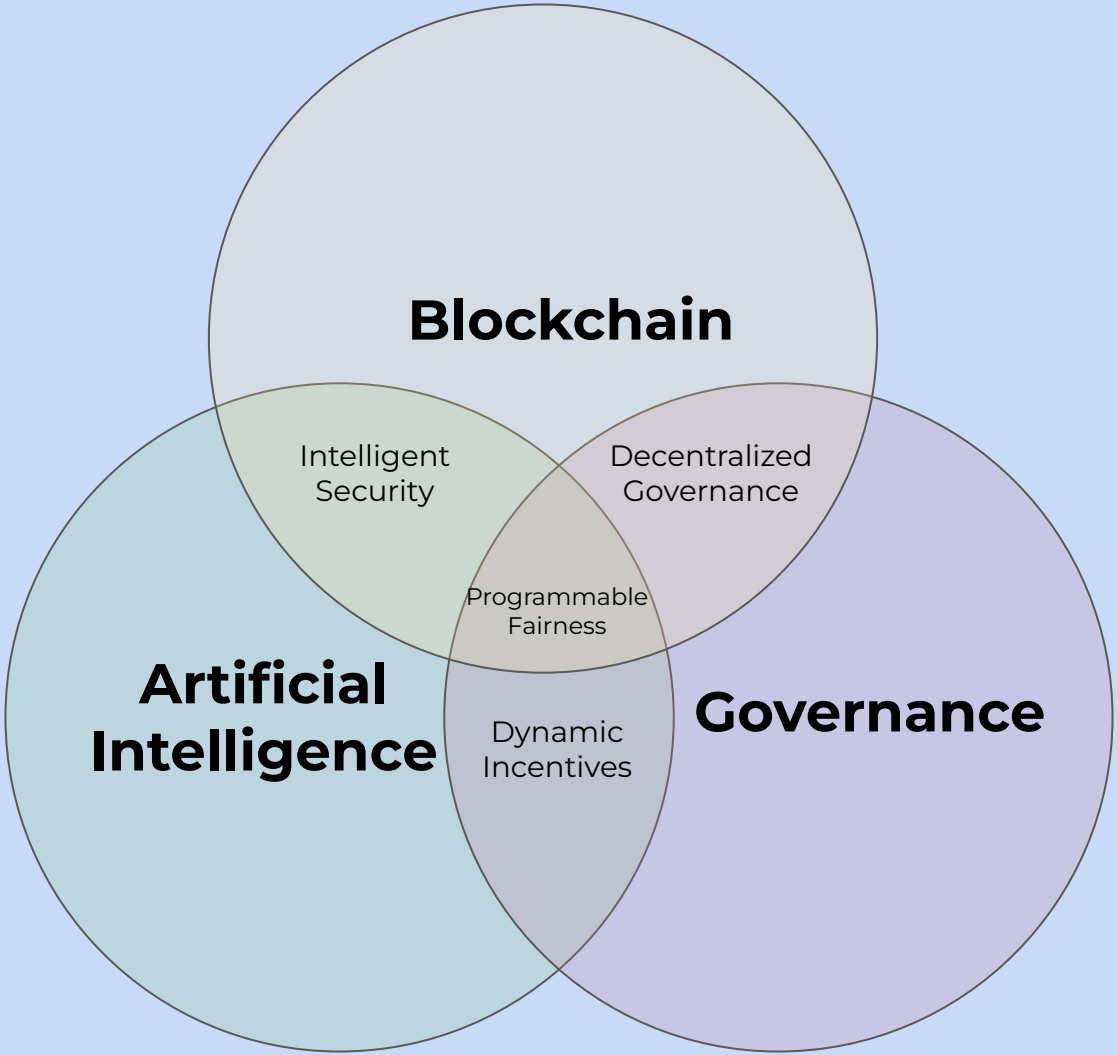
(de)centralization

Regulation

Accountability

Democratization

Transparency


Blockchain
Intelligent Security
Decentralized Governance
Programmable Fairness
Artificial Intelligence
Dynamic Incentives
Governance

Paper Purpose

The purpose of this paper is to examine the significance of artificial intelligence as a novel technological solution to aid in blockchain governance design, specifically token holder voting mechanisms.

1. Briefly introduce the **convergence of artificial intelligence and [blockchain governance]** by focusing on token voting

2. Identify where decision making occurs in [Blockchain Governance] and why we decided to focus on token voting

3. Minimal literature focusing on AI and token voting systems

4. Significant amount of movement towards DAOs and other decentralised governance use cases pursuing AI

5. Decentralised governance is growing at scale, more and more value is locked, more users are participating and even the base technologies are being piloted by real world examples making it a compelling reason to research the topic thoroughly

6. AI is receiving a significant amount of attention

7. Decentralised technologies rapidly evolve, shape faster and adopt new technologies faster than centralised heavy bureaucratic organisation

8. We hope to raise ethical questions about the role of AI in decision-making processes, and provide ground for further analysis

9. Challenge current state of decentralised governance and to offer a window into what the blockchain ecosystem may look like if implementing AI

Dimensions of Blockchain Governance

Definition of Blockchain Governance

"Blockchain governance refers to the structures and processes that are designed to ensure the development and use of blockchain are compliant to legal regulations and ethical responsibilities."
(Liu et al, 2023)

Endogenous Governance
(Proactive)

Exogenous Governance
(Reactive)

liu et al, 2023

Dimensions of Blockchain Governance

*"Blockchain consists of the **three dimensions of decision rights, accountability, and incentives,** whereby they should all align with the decentralisation level of deployed blockchain.*

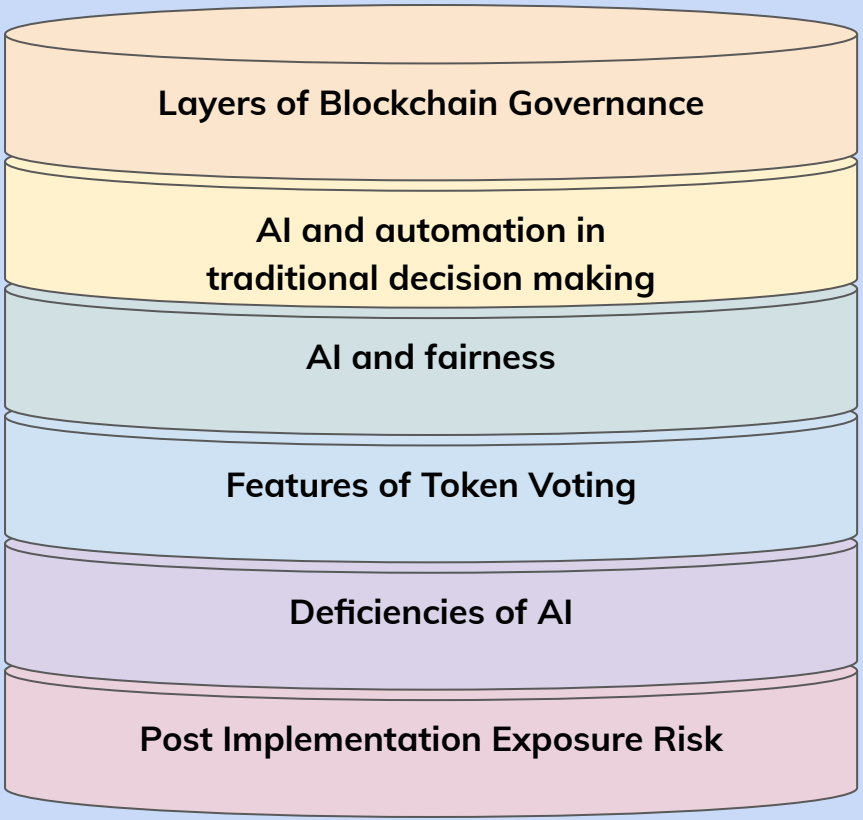
*First, **decision rights of an entity may vary throughout the lifecycle of blockchain.** For example, a user can compete to be a block validator, who is capable to select the transactions for inclusion.*

*Secondly, **pseudonymous address is a challenge** for the identification of accountable entities, particularly in permissionless blockchain.*

*Thirdly, **incentives can be realised** through on-chain token distribution and off-chain business agreements."*

(Liu et al, 2023)

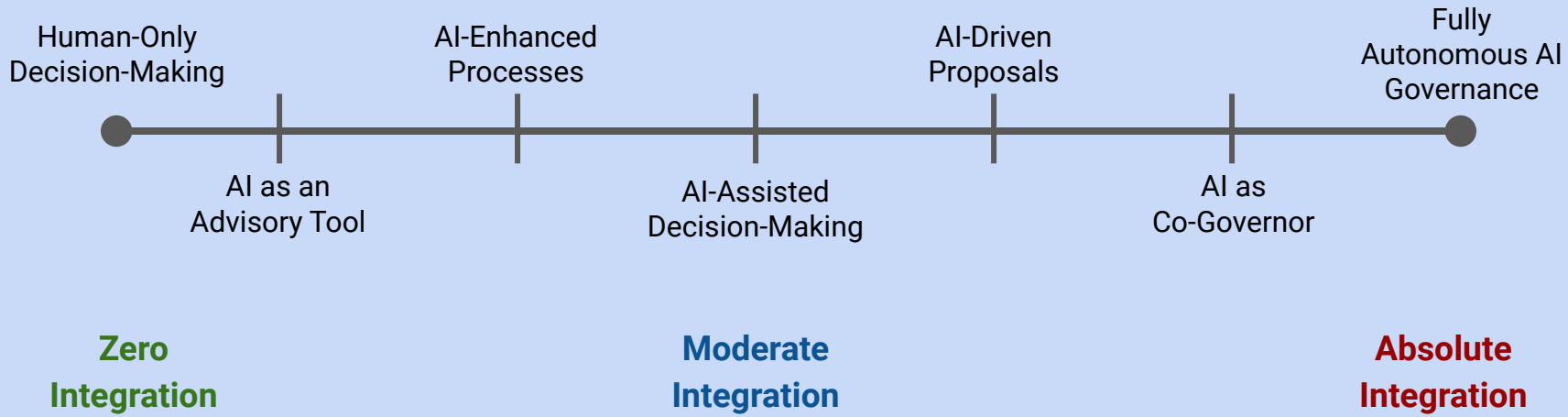
Research Topics



How integrated should AI be in BG?

Endogenous Governance
(Proactive)

Exogenous Governance
(Reactive)



Learnings thus far...part 1

- "A systematic literature review on blockchain governance" (Liu et al, 2023)
- Voter Apathy
- Plutocratic Governance
  - quadratic voting; idea that pro-activity is rewarded (counters voter apathy); increased value with each time you vote)
  - Chainalysis-2022-state-of-web3-report.pdf
- Voter Ignorance
  - AI agent can play critical role;
  - key is training re: pattern recognition/reducing biases; training using supervised/reinforced/unsupervised (where reinforced is guided form of training); reinforced human learning
- Vocal Minority (special interest/lobbying)
- Accountability (balancing w/ privacy)
- Sybil Resistance
- Financial Stratification

Gaps

Problem statement(s):

- Can integrating artificial intelligence enhance fairness and transparency in token voting systems?
    - Current forms of AI require training through data which includes biases from humans and concerns re: quality of data, how does one mitigate these issues to ensure fairness leveraging AI in token voting systems?
    - What areas of token-based governance and decision making can take advantage of AI to improve fairness and transparency?
- Can integrating AI tooling into DAOs mitigate serious voter apathy?
- What forms of AI tools provide advantages to bootstrapping DAOs?
- Where does decision making occur in blockchain governance that rationalises the use of AI tooling?
- What technical infrastructure is required to implement? What are risks associated with implementation?

Future Considerations

Questions:

- Given everything we have discussed, what happens when AI reaches super intelligence?
- Will the “super intelligent AI” be more fair/transparent in the token voting systems?
- Does integration of a “super intelligent” AI mean that DAOs will outlive humans? And is that a positive?
- Is oversight even possible with “super intelligent AI” governance structure?

Interactive Q&A

Questions:

Contact info; telegram:

FASE working group:

Joey Beverley:

T: @joeyNB_COS;

E: Joey.beverley@singularitynet.io

Jessica Mila Schutzman:

T: @jessicamilaschutzman